Corporate Carbon Accounting: A Literature Review of Carbon Accounting Research from the Kyoto Protocol to the Paris Agreement

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Abstract

This paper describes the development of and gaps in knowledge in research on carbon accounting based on a systematic review of 117 papers published in influential accounting journals between 2005 and 2018. The review shows the literature has developed into four major streams of carbon accounting: carbon disclosure, management, performance and assurance and that carbon accounting is emerging as a distinct discipline. Finally, our paper highlights future research opportunities to improve carbon accounting, so it can play an even more important role to help business achieve the climate goals of the Paris Agreement.

Keywords: Carbon accounting, literature review, climate change, corporate carbon management, Paris Agreement

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1. Introduction

Climate change is one of the most serious issues affecting communities and economies around the world.[1] Governments in many countries have designed and implemented various market-based and non-market-based policies to encourage companies to reduce their greenhouse gas (GHG) emissions. Changes in climate conditions and new carbon institutions inevitably influence corporate practices (Howard-Grenville et al., 2014). It is thus not surprising that the interplay among carbon institutions, corporations, and their stakeholders has attracted significant interest from academic researchers, and many papers on carbon accounting have been published in accounting journals.

The objective of the present study is to comprehensively review the research on emerging carbon accounting practice. Our intention is to help researchers understand the current status of the carbon accounting literature and guide future development in this area. Currently there are three reviews of carbon accounting in the accounting literature: Hartmann et al. (2013), Ascui (2014) and Borghei (2021). These reviews have their limitations.[2] First, none of the reviews is comprehensive. Hartmann et al. (2013) only review a few key papers, and their discussion of future research focuses on carbon management and control. Ascui (2014) only covers eight journals, including one non-accounting journal (i.e., the *Journal of Cleaner Production*). Borghei (2021) only focuses on carbon disclosure. Second, Hartmann et al. (2013) and Ascui (2014) only cover papers published in years to 2013. Thus, they miss recent developments in carbon accounting research. In contrast, our review covers a wide range of accounting journals and topics within carbon accounting and includes papers published through the end of 2018. With our comprehensive literature review, we hope to present a full picture of the current carbon accounting research.

For the purpose of this literature review, we adopt the information-perspective definition of carbon accounting proposed by Tang (2017), which defines carbon accounting as "a system that uses accounting methods and procedures to collect, record, and analyse climate change–related information and account for and report carbon-related assets, liabilities, expenses, and income to inform the decision-making processes of internal managers and external stakeholders" (Tang, 2017, p. 11). Based on this definition, we identify carbon accounting publications through a keyword search of the Business Solution Ultimate, ScienceDirect, and Scopus databases. After computer-aided and manual screening, we select

[1] See the United Nations Global Issues website (http://www.un.org/en/sections/issues-depth/global-issues-overview/).

[2] There are literature reviews on carbon accounting in other disciplines. Two examples are Stechemesser and Guenther (2012) and Hahn et al. (2015). The former focuses on defining carbon accounting, and the latter focuses on carbon disclosure.

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117 papers published in quality accounting journals as the source data for this review.

An analysis of publication trends in carbon accounting shows that the initiation and evolution of carbon accounting research and practice is synchronised with the development of global carbon institutions. Early publications in the carbon accounting literature can be traced back to the publication of Freedman and Jaggi (2005), which happened to coincide with the inception of the European Union (EU) Emission Trading Scheme (ETS). Since then, a significant number of conceptual or empirical works have been published in various accounting journals to provide opinions, insights, and evidence on carbon accounting. The development of the carbon accounting research corresponds to the development of global carbon legislation, low-carbon investment, and corporate carbon disclosure.

We find that studies on corporate carbon accounting mainly focus on the following: accounting for carbon assets and liabilities, carbon disclosure, carbon assurance, carbon management, carbon performance, and the impact of carbon issues on the capital market. Of these areas, carbon disclosure is the most extensively studied. Researchers discuss the quality of voluntary carbon disclosure and identify a number of facts external and internal to the firm that affect corporate carbon disclosure. Here we summarise the contributions and limitations of the current carbon accounting research, outline emerging research trends, highlight opportunities to integrate and expand on existing knowledge, as well as discuss avenues for further investigation of corporate carbon accounting.

The remainder of this paper is organised as follows. Section 2 provides an overview of the carbon accounting papers examined and describes publication trends in carbon accounting research in accounting journals. Section 3 provides a detailed review of the carbon accounting research, focusing on corporate practices for dealing with climate change, the drivers behind these practices, as well as their consequences. Section 4 summarises the contributions and limitations of prior carbon accounting studies and suggests future research directions. Section 5 concludes the paper.

2. Method and descriptive analysis

2.1 Literature search

The literature search involves three main steps. First, a general keyword search is performed in three major journal databases: Business Solution Ultimate, ScienceDirect, and Scopus. The keywords are selected based on the definition of carbon accounting discussed earlier to reflect the two essential elements of carbon accounting research: it relates to climate change; and it relates to management, accounting, accountability, or stakeholders. Thus, we

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choose two groups of keywords. The first group includes seven keywords related to climate change: *climate change*, *global warming*, *carbon*, *greenhouse*, *GHG*, *emission**, and *ETS*. The second group includes eight business-related keywords: *account**, *manage**, *finance**, *assur**, *audit**, *disclos**, *report**, and *valu**. The search is based on 56 possible phrases, each one combining one climate change keyword and one business keyword with the Boolean operator *AND* in between. We limit keyword matching to the abstract, title, or keyword if the database provides this functionality; otherwise, keyword matching is based on the full text. The search results are imported into EndNote, and duplicate papers are deleted. The search returns 11,112 unique records from 1,372 journals. These journals cover a wide range of research disciplines, including accounting, business, economics, science, engineering, politics, and law.

Second, the search results are screened by journal. To include only papers published in high-quality accounting journals, we rely on the Australian Business Dean Council (ABDC) journal list for this screening. The ABDC is the "collective voice of university pro vice-chancellors, executive deans, and heads of Australian business faculties and schools" representing 39 university faculties and schools in Australia. The council publishes a journal quality list that rates more than 2,700 research journals across 16 business-related areas as A*, A, B, or C, with A* representing journals of the highest quality. This list is commonly used by member universities as a benchmark for evaluating the research performance of their staff. This review focuses on journals with an A*, A, or B rating, which can be largely matched to Association of Business School Academic Journal Guide 2015 (ABS) rankings 4, 3, and 2, respectively.[3] This screening process narrows the sample to 621 papers published in accounting journals.

Third, a final manual screen is conducted to exclude papers with little relevance to corporate carbon accounting. Specifically, we exclude papers that focus on issues other than climate change as well as papers that are related to climate change but irrelevant to corporations. The final sample consists of 117 papers on carbon accounting published in 34 influential accounting journals from 2005 to December 2018.[4]

2.2 Descriptive analysis

Table 1 presents the list of journals in which the selected papers are published. The table organises the journals according to their ABDC rating and also shows their corresponding

[3]For more information on the review process for the ABDC journal list, see http://www.abdc.edu.au/pages/2016-review.html.

[4] Some carbon accounting papers published in non-accounting journals are identified in the search process. For example, through the end of 2018, 20 papers are published in the *Journal of Business Ethics* and four papers are published in the *Journal of Cleaner Production*. These papers concern carbon disclosure, carbon risk management, supply chain carbon management, and carbon strategy and management. Although these papers are not included in this review, we acknowledge their contribution to the literature.

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Association of Business School Academic Journal Guide 2015 ranking. In the A* group (2018 rating), *Accounting, Organizations and Society* has the most carbon accounting papers (5 papers), followed by *European Accounting Review* (4). In the A group, *Accounting, Auditing and Accountability Journal* has the most papers (18). It is also the journal with the most publications across the entire sample. *British Accounting Review* ranks second (8) in this group in terms of the number of publications. In the B group, *Australian Accounting Review* and *Sustainability Accounting, Management and Policy Journal* are on top with 10 and 7 papers, respectively. On average, each journal publishes 3.44 papers (117/34). The average number of publications is 3 (15/5) in A* journals, 3.73 (56/15) in A journals, and 3.28 (46/14) in B journals.

<Insert Table 1 here>

Table 2 summarises the distribution of publications over the period 2005–2018, and Figure 1 visualises the trend in publication. The number of publications on carbon accounting increases sharply from 2005 to 2011 and then becomes stable from 2012 to 2018. Next, we classify the publications by methodology: conceptual, qualitative empirical, or quantitative empirical. Table 3 and Figure 2 show the distribution of papers across these three categories.

<Insert Tables 2 and 3 here>

<Insert Figures 1 and 2 here>

The emergence and development of carbon accounting research represents the response of accounting academics to the challenge of climate change. Figure 3 illustrates how this response resonates and corresponds with the responses of other groups and communities in society to climate change. In particular, similar increasing patterns exist in the amount of carbon legislation enacted, the amount of green investment, the number of companies that voluntarily participate in the CDP survey, and the number of publications on carbon accounting from 2004 to 2018. These patterns reflect the concurrent activity of legislators, investors, corporations, and academics, respectively. It appears that social actors respond to climate change in an interactive manner and reinforce one another.

<Insert Figure 3 here>

Next, we examine the shift in methodology in the carbon accounting literature. Conceptual papers are dominant in the early development of the literature until 2011. Empirical studies gain in popularity beginning in 2012. In particular, the number of quantitative studies increases steadily beginning in 2014, possibly owing to the increased accessibility and availability of carbon-related data.

In sum, our descriptive analysis of the papers identifies a few features of the carbon

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accounting literature. First, carbon accounting studies are published in a wide range of accounting journals. Second, the increase in the publication of carbon accounting papers in accounting journals is synchronised with trends in carbon legislation, green investment, and corporate carbon disclosure. Finally, the methodology used in carbon accounting research has largely shifted from conceptual to empirical study, especially quantitative empirical study.

3. The carbon accounting literature

The Kyoto Protocol established the carbon market as a market-based mechanism for reducing or limiting GHG emissions in ratifying countries. The formation of the carbon emissions trading market created accounting issues that had never before been considered. The new mechanism aroused critical discussion and debate in the accounting literature on the nature and effectiveness of various climate change policies (Andrew & Cortese, 2013; Andrew et al., 2010; Boston & Lempp, 2011; Callon, 2009; Carter et al., 2011; Lodhia & Martin, 2012a; McNicholas & Windsor, 2011). For example, carbon markets are considered "on-going collective experiments" that redefine "the forms of organization of economic, political and scientific activities" (Callon, 2009, p. 549). Moreover, the process of setting climate change policy is regarded as highly political (Andrew & Cortese, 2013; Boston & Lempp, 2011; Carter et al., 2011), and doubt has been cast on the effectiveness of various climate change policies for reducing GHG emissions (Andrew et al., 2010; Callon, 2009; McNicholas & Windsor, 2011).

Although the policy debate considers the impact of climate change on business at the macrolevel, some scholars seek to identify the specific accounting implications of climate change and discuss the challenges to carbon accounting in terms of both research and practice (e.g. Bebbington & Larrinaga-González, 2008; Hopwood, 2009) _ENREF_84. In particular, they discuss financial accounting and reporting issues related to carbon trading as well as ways in which accounting may be involved in communicating climate risks and uncertainties. Hartmann et al. (2013) extend this discussion and explicate carbon accounting from a management accounting perspective to promote research in carbon management and control. These early conceptual studies outline the opportunities and challenges for later carbon accounting research.

In the rest of the section, we review the literature related to corporate carbon accounting. This literature is organised according to the following subject matter: financial accounting for carbon assets and liabilities, carbon disclosure, carbon assurance, carbon management, carbon performance, and the impact of carbon issues on the capital market. A list of the papers in each category is presented in Table 4.

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3.1 Financial accounting for carbon assets and liabilities

The introduction of the carbon trading market raises a typical financial accounting question—that is, how carbon allowances should be recorded in the books. The debate focuses on the valuation of allowances allocated for free and the volatility caused by different valuations of assets and recognition of liabilities (Bebbington & Larrinaga-González, 2008). The difficulty has to do mainly with the multiple implications of carbon allowances. Specifically, a carbon allowance acts as a permit for carbon emissions, a mandatory upper limit of carbon emissions, and a financial asset that can be traded on the market. The conflict created by this unique feature of carbon allowances and the need for consistency between carbon-related assets/liabilities and other general assets/liabilities makes it difficult for standard setters to design a universal accounting standard (Cook, 2009).

Disagreement over accounting for carbon allowances led to the withdrawal of International Financial Reporting Interpretations Committee Interpretation 3: Emission Rights in June 2005. In the absence of accounting standards for emissions allowances, three different approaches are commonly used in practice: (1) a net liability approach that classifies allowances as intangible assets and only shows an emissions liability when emissions exceed free allocated allowances, (2) a gross liability approach that recognises the free allocation at fair value and a corresponding gross liability, and (3) an inventory approach with free allocations given at nil value (Black, 2013). A major concern around the discrepancies in accounting is that they can hinder the comparability of firms' financial statements (Griffin, 2013; Warwick & Ng, 2012) because firms have total discretion over the recognition and measurement of carbon allowances as either assets, liabilities, or even expenses (Mete et al., 2010). Therefore, scholars call for a common approach to be adopted by accounting standard setters (Giner, 2014; Haupt & Ismer, 2013; Raibor & Massoud, 2010).[5]

Discussion of accounting for carbon allowances has faded out of accounting publications since 2014. However, the issue still exists and may become more significant when more and more countries around the world start to adopt ETSs as means of controlling GHG emissions. Therefore, researchers can contribute to research on carbon financial accounting by further investigating this issue. Moreover, the treatment of carbon allowances is not the only issue in financial accounting related to climate change. Ratnatunga et al. (2011) propose their own model of accounting for various carbon-related items, such as emissions allowances (or

[5] This issue is still on standard setters' agenda. In December 2012, the International Accounting Standards Board formally reactivated the project to consider the accounting treatment of carbon emission allowances awarded by a scheme administrator and liabilities associated with the emission of GHGs. In 2015, the project was re-scoped to consider a variety of schemes that use emissions allowances to manage the emission of pollutants. No further update after October 2015 can be found on the Board's website (see https://www.iasplus.com/en/projects/research/short-term/emissions-trading).

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rights), carbon sequestration, and emissions control capabilities. Other items worthy of investigation include stranded assets and emissions control assets. We believe that further incorporation of carbon issues into accounting will be a fruitful avenue of research when the impact of climate change on the value of corporate assets becomes more significant.

3.2 Carbon disclosure

Corporations make disclosures around climate change through various channels. Some companies report GHG emissions to government agencies under certain regulatory schemes, others disclose climate change information in their annual reports or in stand-alone sustainability or corporate social responsibility (CSR) reports, and still others voluntarily make more detailed disclosures through the CDP (Huggins et al., 2011). Of all carbon accounting issues, carbon disclosure is the most extensively studied (see Table 4). Studies on carbon disclosure can generally be divided into two categories. Studies in the first category are concerned with external and internal factors that affect firms' carbon disclosure decisions, and studies in the second evaluate and comment on the quality and adequacy of voluntary carbon disclosure.

3.2.1 Determinants and motivations of carbon disclosure

Research on the determinants of carbon disclosure helps researchers and policymakers understand how firms interpret and respond to pressures imposed by governments, communities, and external groups and may assist in establishing effective regulatory frameworks for promoting transparency in carbon control. A few theories on voluntary carbon disclosure, namely, legitimacy theory, stakeholder theory, signalling theory, and institutional theory, are usually used as underlying theories in investigations of determinants and motivations of carbon disclosure. Although the theories take different perspectives, in general they suggest that a firm's propensity to disclose its true position regarding carbon emissions and carbon management is associated with social, market, economic, regulatory, and institutional pressures when they are transferred into disclosure incentives. This relationship is illustrated in Figure 4, which is adapted from Luo et al. (2013).

<Insert Figure 4 here>

External pressures

Part of the literature considers how regulatory and institutional pressures affect firms' carbon disclosure. Researchers find that legal systems with strong protection of investors and stringent environmental regulations (such as common law systems) drive voluntary carbon disclosure (Alrazi et al., 2016; Luo et al., 2012; Tang & Luo, 2016). Another finding is that firms in carbon-intensive sectors tend to make more carbon disclosures than firms in other

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sectors because they face higher regulatory pressure (Choi et al., 2013; Chu et al., 2013; Rankin et al., 2011; Tang & Luo, 2016). The positive influence of governmental or stock exchange regulations around carbon disclosure on the tendency to voluntarily disclose carbon information is confirmed by empirical studies set in Australia (Cowan & Deegan, 2011; Liu et al., 2017), the United Kingdom (de Aguiar & Bebbington, 2014; Tauringana & Chithambo, 2015), the United States (Freedman & Park, 2014), and China (Yang & Farley, 2016). Furthermore, guidance from nongovernmental institutions, such as the Global Reporting Initiative and CDP, can also positively influence the extensiveness and credibility of corporate carbon disclosure (Rankin et al., 2011).

Some other studies consider the role of economic pressure in influencing carbon disclosure. One source of economic pressure is internalised carbon-related costs due to the implementation of climate policy (Bebbington & Larrinaga-González, 2008). Freedman and Jaggi (2005) show that firms with headquarters in countries that have ratified the Kyoto Protocol are more likely to provide detailed disclosures of their efforts and achievements when they are required to make large expenditures to meet the requirements of the Protocol.[6]Another source of economic pressure is a firm's operating environment. Ott et al. (2017) argue that competition pressure may affect firms' decisions to voluntarily disclose carbon activity. They find that firms in more concentrated markets, in markets with more substitutability, and in smaller markets are more likely to make their responses to the CDP survey publicly available because of increased pressure on firm operating efficiency and lower entry barriers to entry the markets.

Other external pressures that have been considered in the literature include social pressures and financial market pressures. Research generally uses firm size as a proxy for social pressures, because it can be argued that larger firms attract greater attention from the media and regulators. In different settings, researchers confirm that social pressure is one driver of the likelihood and extent of voluntary carbon disclosure (Choi et al., 2013; Chu et al., 2013; Freedman & Jaggi, 2005; Luo et al., 2012)_ENREF_30. Regarding financial market pressure, it is expected that investors play a crucial role in driving managers to disclose more information on GHG emissions in an effort to understand the exposure of their investment to climate risks and opportunities. However, empirical findings generally suggest no link or even a negative relationship between capital raising (Luo et al., 2012; Tang & Luo, 2016)_ENREF_144 or ownership concentration (Liao et al., 2015; Liesen et al., 2015; Tauringana & Chithambo, 2015)_ENREF_94_ENREF_96 and public carbon disclosure.

[6] Later studies cannot confirm their findings, possibly because the vast majority of countries have ratified the Protocol (see Alrazi et al., 2016; Luo et al., 2012; Tang & Luo, 2016).

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Firm characteristics

One firm characteristic that has been examined as a determinant of carbon disclosure is corporate governance structure. Two Australian studies, Rankin et al. (2011) and Choi et al. (2013), find a positive association between the overall quality of corporate governance and the propensity and extensiveness of carbon disclosure. Regarding specific board characteristics, Liao et al. (2015), Elsayih et al. (2018), and Hollindale et al. (2019) show that the percentage of female directors on the board is positively associated with the propensity to disclose carbon information. Liao et al. (2015) and Elsayih et al. (2018) further show that independent directors tend to favour more transparent carbon disclosure owing to their diverse backgrounds and lack of financial stake in the firm. Tauringana and Chithambo (2015) find that director ownership is negatively associated with a firm's carbon disclosure, which suggests that directors with higher ownership may resist undertaking carbon abatement activities, including disclosure, because of the uncertainty of a payback on such a substantial environmental investment. Conversely, Elsayih et al. (2018) find a positive correlation between management ownership and carbon disclosure. They argue that enhanced management ownership may help align the interests of managers and shareholders; hence, managers would be more willing to communicate with shareholders with regard to climate change.

Other internal factors considered in the literature are financial characteristics, specifically, profitability, leverage, and growth opportunities. Researchers generally expect a positive association between firm profitability and carbon disclosure (e.g. Choi et al., 2013; Chu et al., 2013; Freedman & Jaggi, 2005; Luo et al., 2013; Ott et al., 2017) based on the argument that higher profitability increases a firm's available financial resources and allows the firm to invest in carbon abatement and carbon disclosure. However, of all of the studies we review, only Ott et al. (2017) find a positive association between firms' decision to respond to the CDP survey and return on assets. All other studies fail to find significant effects of firm profitability on either the propensity or extent of carbon disclosure. Regarding the effects of leverage, it can be argued that higher leverage indicates more pressure, so management would be expected to disclose carbon information (Tang & Luo, 2016). However, a highly leveraged firm may prefer to withhold sensitive carbon risk information, fearing that heightened transparency could undermine its negotiating position (Luo & Tang, 2014b). The empirical results are inconclusive, with some studies finding a positive association (Tang & Luo, 2016), some a negative association (Luo & Tang, 2014b), and most no association (Freedman & Jaggi, 2005; Liesen et al., 2015; Luo et al., 2012; Luo & Tang, 2014b)_ENREF_96_ENREF_53_ENREF_104. In terms of firm growth opportunities, Luo et al. (2013) show that firms with high growth opportunities are less likely to make carbon disclosures, especially in less advanced economies. This finding indicates that when firms

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prioritise financial goals over environmental concerns, they divert fewer resources to carbon reduction and disclosure.

Still other factors that may affect a firm's decision to disclose carbon information include managers' attitudes and philosophies around environmental protection. Luo and Tang (2016b) examine the association between national culture and firms' responses to the CDP survey, given that culture can significantly influence managers' responses to carbon issues. They find that the cultural dimensions of masculinity, power distance, uncertainty avoidance, individualism, and long-term orientation are significantly related to the propensity to disclose carbon information. Finally, the presence of an environmental management system in a firm is positively associated with the propensity to disclose as well as the credibility and extent of such disclosure, either because firms with an environmental management system are more likely to communicate their efforts for dealing with climate change to powerful stakeholders such as investors and the general public (Rankin et al. (2011) or because they have considerably lower additional preparation costs for public carbon disclosure (Ott et al. (2017).

3.2.2 The quality and adequacy of carbon disclosure

As corporate carbon disclosure remains voluntary in most of the world, several researchers express concerns about the quality of corporate carbon disclosure (e.g. Haque & Deegan, 2010; Kolk et al., 2008; Stanny, 2018). More specifically, Cotter et al. (2011) find evidence that corporate disclosures tend to lack technical details and are somewhat skewed toward more positive aspects of climate change management. Comyns and Figge (2015) measure disclosure quality with a self-constructed index based on seven principles: accuracy, completeness, consistency, credibility, relevance, timeliness, and transparency. They find that disclosure quality has not improved significantly and suggest that regulation is needed to discipline the practice. Haslam et al. (2014) criticise existing approaches that result in disclosures that "generate malleable, inconsistent and irreconcilable numbers and narratives." Haigh and Shapiro (2012) find that carbon emissions reports have some use in investors' assessments of firms' corporate governance but that the information on carbon performance may not affect investors' portfolio decisions. Solomon et al. (2011) explore the nature of one-on-one private meetings between institutional investors and their investee companies. The authors find that investors use the private reporting process to compensate for the acknowledged inadequacies of public climate change reporting.

The inadequacy of corporate carbon disclosure reflects the gap in expectations between stakeholders and corporate managers. While green communities and environmental regulators concern themselves with carbon pollution, managers focus primarily on compliance costs and risks and on financial performance for shareholders (Haque et al., 2016; Lodhia & Martin,

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2012b) and do not appear to be held accountable for the negative impact of operations on climate change (Haque et al., 2016). Furthermore, methodological heterogeneity due to the voluntary nature of carbon disclosures may result in incomparable data and undermine the usefulness of such information; hence, mandatory regulation might be a solution (Andrew & Cortese, 2011). Generally speaking, concerns over the quality and reliability of voluntary carbon disclosure in the carbon accounting literature resonate with the critiques of early authors of sustainability accounting and disclosure (Gray, 2010; Hopwood, 2009). However, some authors point out that disclosure under the CDP system has improved (e.g. Luo & Tang, 2014b) and that carbon disclosure helps investors make decisions regarding firm market valuation (Matsumura et al., 2014).

3.2.3 The relationship between carbon performance and carbon disclosure

Concerns over the quality of voluntary carbon disclosure raise the question of whether the carbon information voluntarily disclosed reflects a firm's underlying carbon performance, because there must be correspondence between the carbon disclosure and the firm's actual carbon performance for the disclosure to be useful. Two competing theories, signalling theory and legitimacy theory, are often proposed to explain the association between carbon performance and disclosure. Signalling theory suggests that adverse selection encourages good performers to make more credible, objective, and quantitative disclosures so that they can distinguish themselves from poor performers (Spence, 1973; Verrecchia, 1983). Conversely, legitimacy theory suggests that the legitimacy threat motivates poor performers to disclose more (Aerts & Cormier, 2009; Gray et al., 1995; Neu et al., 1998).

Unfortunately, the empirical results for this research question are not conclusive. Luo and Tang (2014b) and Datt et al. (2019b) find support for signalling theory by showing that the extensiveness of carbon disclosure is positively associated with the firm's carbon mitigation performance. Ott et al. (2017) present similar but relatively weak evidence. In contrast, Momin et al. (2017), using data from China, find evidence consistent with legitimacy theory, whereas Alrazi et al. (2016) find no significant results. Luo (2019) offers another perspective by explaining that stringent carbon institutions can restrict firms' incentives to use carbon disclosure for legitimation attempts. Furthermore, Hrasky (2012) points out that whereas firms in carbon-intensive sectors reveal more behavioural information to signal their substantive action in coping with climate change, those in less carbon-intensive sectors rely more heavily on disclosing symbolic information. Ferguson et al. (2016) contend that the disclosures produced by participants in the UK ETS and Energy Efficiency Scheme not only were used to lend legitimacy to the organisations but also served to displace their own responsibility for addressing climate change and shift blame onto the government or suppliers as barriers to

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progress.

While firms' carbon performance may influence their disclosure behaviour, carbon disclosure impacts their carbon performance. Qian and Schaltegger (2017) argue that no matter whether firms make carbon disclosures for legitimacy or signalling purposes, carbon disclosure motivates companies and drives subsequent organisational change from the outside in. Their results show that change in carbon disclosure levels is positively associated with change in subsequent carbon performance.

3.2.4 Summary

Companies disclose carbon-related information through annual reports, sustainability or CSR reports, company websites, and third-party channels such as the CDP survey. To date carbon disclosure remains largely voluntary, and companies have discretion in whether, what, and how much to disclose. The voluntary nature of carbon disclosure naturally stimulates two research questions. First, what factors affect firms' carbon disclosure decisions in terms of choice, content, and extent? Second, how good are voluntary carbon disclosures? By reviewing the literature, we find that these two questions have received extensive examination.

In terms of the first question, the literature identifies a number of country-, sector-, and firm-level factors related to firms' carbon disclosure decisions. We believe that this line of enquiry leaves much room for exploration. For example, although climate change mitigation is a highly political issue, prior studies have not considered politics as a country-level factor that may be associated with carbon disclosure. Researchers can also refer to our framework of determinants of carbon disclosure to identify other gaps in this area. The second question points out that voluntary carbon disclosure as it stands currently may be of low quality and inadequate. As suggested by Comyns and Figge (2015), disclosure quality is a multidimensional construct that includes accuracy, completeness, consistency, credibility, relevance, timeliness, and transparency. The current literature largely focuses on the completeness and consistency of disclosures (Cotter et al., 2011; Haslam et al., 2014). Future research can look more closely at other dimensions of quality. _ENREF_74A common issue related to disclosure quality is greenwashing (i.e., companies using voluntary environmental disclosure to portray themselves as ecologically friendly when they are not; (Delmas & Burbano, 2011). Thus, sustainability reporting could actually reduce what is known about a company and act as a corporate veil (Hopwood, 2009). Carbon accounting researchers can contribute to the literature by designing a systemic approach to detecting greenwashing and examining the causes and consequences of this practice. Furthermore, the research on carbon disclosure should certainly not stop at the two aforementioned research questions. Researchers can also explore the information needs of stakeholders to guide company practices and investigate the social and economic consequences

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of carbon disclosure.

3.3 Carbon assurance

The assurance of GHG statements is an emerging practice developed in response to evolving carbon regulations, which require accurate and reliable information on GHG emissions (Martinov-Bennie, 2012; Martinov-Bennie & Hoffman, 2012; Simnett et al., 2009). Carbon assurance is different from traditional financial auditing in terms of the legal and regulatory environment, organisational participation, the materiality threshold, competencies required, accounting methods, and users of the assurance reports (Olson, 2010). The current limited literature on carbon assurance mainly focuses on carbon assurance practices and determinants of firms' carbon assurance choices.

Green and Zhou (2013) and Datt et al. (2018) provide detailed examinations of carbon assurance practices around the world. Their findings reflect the growth and evolving nature of this emerging market. Because carbon assurance requires not only auditing expertise but specific subject matter expertise related to GHG emissions (Green & Taylor, 2013; Huggins et al., 2011; Olson, 2010), two major types of carbon assurance provider exist: accounting firms and consultation firms. Although they perform the same task, they vary in terms of expertise required, skill, and independence (Kim et al., 2016). One popular topic in the assurance practice research is that of expectation gaps. Survey results from Green and Taylor (2013) and Green and Li (2012) indicate that expectation gaps exist among different groups of stakeholders in regard to the perceived quality of the assurance provider and the responsibilities of assurers and managers.

Current research on the determinants of voluntary carbon assurance considers the choice to engage and choice of assurance provider. Zhou et al. (2016) consider the influence of institutional background on firms' carbon assurance choices. They find that firms in stakeholder-oriented countries and countries with weaker legal enforcement are more likely to seek out carbon assurance. These factors also affect firms' choice of carbon assurance provider. Datt et al. (2019a) find that firms with a greater legitimacy threat are more likely to seek out carbon assurance, which indicates that external carbon assurance plays a role in enhancing the credibility of carbon disclosure and mitigating legitimacy threats. Datt et al. (2018) take a multitheoretical perspective and find that carbon risks, carbon reduction initiatives, environmental committee, staff involvement in carbon actions, and carbon disclosure scores are positively associated with a firm's propensity to obtain assurance. Furthermore, Tang (2019) shows that the creation of carbon institutions and significant increases in governmental green funding are the main drivers of the emergence and development of carbon auditing in a Chinese setting. He argues that carbon auditing has been used as a tool of social and economic reform

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in China for the promotion of sustainable sociotechnical and organisational innovation.

In sum, the literature considers incentives and determinants of the choice to seek out voluntary carbon assurance as well as the choice of assurance provider (Datt et al., 2019a; Datt et al., 2018; Zhou et al., 2016). Another focus is expectation gaps between users, assurers, and companies regarding the purpose, responsibility, and quality of carbon assurance (Green & Li, 2012; Green & Taylor, 2013). Despite the insights provided by the research, understanding of carbon assurance practices is still very limited. In particular, knowledge of the methodology of carbon assurance, including internal control, risk assessment, and investigation, is absent from the current literature. Therefore, in-depth research on the methodology of carbon assurance is urgently needed. Furthermore, the research on current determinants focuses only on the choice to engage and the choice of provider; further research can investigate other choices, for example, the scale (scope 1, scope 2, or scope 3 emissions), level (percentage of carbon emissions assured), or type (limited, reasonable, or high assurance) of assurance. Another promising area of future research is the quality of carbon assurance. Audit quality has always been a hot topic in the financial auditing literature (DeFond & Zhang, 2014; Watkins et al., 2004). However, the literature is silent on the quality of carbon assurance.

3.4 Carbon management

A narrow definition of carbon management refers to the practices a company undertakes to mitigate its operational GHG emissions (Hartmann et al., 2013). However, this concept can be extended to the management of all carbon-related issues of the company (Tang & Luo, 2014). In this section we review studies related to carbon management. Generally speaking, the literature on carbon management can be divided into two categories. The first category includes descriptive studies that report carbon management practices and organisational changes related to these practices, and the second includes studies that examine factors that affect firms' carbon management practices.

Firms use various carbon accounting techniques to help managers collect and record carbon-related information (Burritt et al., 2011; Gibassier & Schaltegger, 2015), evaluate performance (Lee & Wu, 2014)_ENREF_52, perform green investment and project management (Vesty et al. (2015), implement carbon strategies (Bui & Fowler, 2019), conduct emissions trading (Engels (2009), manage exposure to risk (Kumarasiri & Gunasekarage, 2017), control carbon costs (Ratnatunga & Balachandran, 2009), and assist in adapting to climate change (Linnenluecke et al. (2015). Studies highlight the importance of embedding climate change information in carbon accounting practices. In addition, introducing carbon management accounting into an organisation may change the work and behaviour of actors and the way they react to authority and climate change (Vesty et al., 2015). Furthermore, Tang and

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Luo (2014) propose a carbon management system that covers four broad components of carbon management: carbon governance, carbon operation, performance tracking, and stakeholder engagement. A carbon management system is a functional tool used by a firm to implement its carbon policy, manage carbon risks, and efficiently mitigate carbon emissions.

Regarding the factors that affect firm carbon management practices, prior studies show that carbon regulation/institution and regulatory uncertainty are closely related to firms' GHG control (Moore & McPhail, 2016). Specifically, Kumarasiri and Jubb (2016) find that regulations that impact companies' economic interests are likely to drive the use of carbon management accounting techniques such as target setting, performance measurement, and incentives for emissions mitigation. Similarly, Bui et al. (2020) find that firms under regulatory pressure tend to use tight budgetary control in carbon management. Another driver of proactive carbon management strategies is a firm's carbon exposure (Jeffrey & Perkins, 2014; Luo & Tang, 2016a). For example, managers' perceptions of climate change risks and opportunities motivate the use of carbon management accounting and reporting (Herbohn et al., 2012; Trotman & Trotman, 2015). Other influential factors identified in the literature include reputational concerns and protection of economic interests (Kumarasiri & Gunasekarage, 2017; Trotman & Trotman, 2015), carbon cost structures (Cadez & Guilding, 2017), management remuneration plans (Deegan and Islam (2012), and corporate governance structure (Yunus et al., 2016).

The carbon management accounting literature enhances understanding of how accounting can help with carbon management. Many accounting concepts and approaches, such as reporting procedures, performance evaluation, risk assessment, and target setting, are embedded in carbon management systems that enhance firms' capabilities for dealing with climate change (Tang & Luo, 2014). Researchers can make in-depth enquiries into each type of carbon management tool. An example of this type of research is Ioannou et al. (2016), who investigate the use of target setting in carbon management. They find that the difficulty of the carbon reduction target is positively associated with the percentage of target completion and that this effect is negatively moderated by the provision of monetary incentives.

We also suggest that carbon accounting researchers pay attention to carbon risk management. Although this topic is rarely addressed in accounting journals, some research on carbon risk has emerged in other business journals. Studies describe the carbon risks to which firms are exposed (Sakhel, 2017), factors that affect firm carbon risk management practices (Elijido-Ten & Clarkson, 2017; Subramaniam et al., 2015)_ENREF_45, and the potential impact of carbon risk on firm capital financing (Herbohn et al., 2019; Jung et al., 2018) and decision making (Haney, 2017).

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Another area worthy of investigation is supply chain carbon management, which involves a comprehensive, cradle-to-grave analysis of the emissions of a product to achieve overall carbon mitigation or reduction (Butner et al., 2008). The carbon accounting literature remains silent on this topic. Future research may pay attention to operational issues related to the implementation of this approach. Potential research avenues include a determination of carbon emissions in the supply chain, an assessment of the carbon performance of each entity in the supply chain, and an investigation into uncertainty among supply chain partners around sharing sensitive proprietary information. Furthermore, accounting for carbon control in the supply chain goes beyond the scope of traditional financial accounting because it involves multiple reporting entities. Some enlightening accounting methods may need to be adopted to address the control of carbon emissions in the supply chain setting.

3.5 Carbon performance

Carbon performance generally refers to how well firms do at managing and controlling carbon emissions. The literature on carbon performance has two main focuses: the complexity of measuring carbon performance and the factors that affect firms' carbon performance.

The concept of carbon performance may be more complicated than it looks. First, measuring GHG emissions is a complex process and is largely based on estimation. The reliability and comparability of the results are questionable when different methods of estimating are used (MacKenzie, 2009). Second, researchers who measure emissions often encounter issues such as boundaries, scope, and aggregate basis (Bowen & Wittneben, 2011; Cooper & Pearce, 2011; Milne & Grubnic, 2011; Young, 2010)_ENREF_111. Third, the term *carbon performance* can have different meanings depending on the purpose of the performance evaluation. It can be an absolute measure of GHG emissions, it can be an efficiency measure that equals GHG emissions per unit of production (such as gross domestic product or sales revenue), or it can reflect the change in the aforementioned measures across two time periods. Some studies also suggest that carbon abatement initiatives can be deemed process-based measures of carbon performance (Busch & Hoffmann, 2011; Haque, 2017). In empirical research, the first two concerns are generally ignored. As for the third, researchers use various measures of carbon performance depending on their research question. For example, Luo and Tang (2014b) suggest that the amount of absolute carbon emissions and carbon intensity are the most appropriate indicators, as the ultimate objective of carbon management is to reduce carbon emissions. Qian and Schaltegger (2017), in contrast, use the change in the intensity of direct and indirect carbon emissions as a measure of carbon performance.

With regard to factors that affect firm carbon performance, Tang and Luo (2014) demonstrate a significant impact of the overall quality of the carbon management system on

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the carbon performance of Australian firms. Jeffrey and Perkins (2015) show the positive effect of energy tax on reducing carbon intensity. Birchall et al. (2015) suggest that carbon accounting is not necessarily evidence of improvement in climate change abatement and that the voluntary carbon market is only moderately effective at decreasing emissions. Haque (2017) finds that certain corporate governance mechanisms are associated with process-based carbon performance (the undertaking of carbon mitigation initiatives) but not with outcome-based carbon performance (carbon emissions). Finally, Broadstock et al. (2018) demonstrate that carbon performance has a nonlinear relationship with firm financial performance, which indicates that the existence of a possible win–win situation is conditional.

Research on carbon performance shows that the complexity involved in measuring carbon performance may limit the reliability and comparability of carbon emissions reporting. Therefore, caution should be taken when interpreting and using self-reported emissions data. Furthermore, our review shows that carbon performance as a research topic is still underexplored, although some studies have investigated the value relevance of carbon emissions, which we discuss next.

3.6 The impact of carbon issues on the capital market

Prior research examines investor reactions to GHG emissions and corporate efforts at carbon mitigation. It can be argued that carbon emissions diminish firm market value—the underlying rationale for this argument is that present and future compliance and carbon mitigation costs represent unbooked liabilities for firms. Studies that directly investigate the valuation effects of carbon emissions in the United Kingdom, the EU, the United States, and Japan (Baboukardos, 2017; Clarkson et al., 2015; Griffin et al., 2017; Johnston et al., 2008; Matsumura et al., 2014; Saka & Oshika, 2014) provide consistent evidence in support of this argument. In addition, it appears that carbon disclosure might, to some degree, moderate the negative effects of carbon emissions on firm value (Baboukardos, 2017; Griffin & Sun, 2013; Liesen et al., 2017; Matsumura et al., 2014; Saka & Oshika, 2014).

Looking beyond the general valuation effects of carbon emissions, Clarkson et al. (2015) contend that not all emissions are relevant to valuation (i.e., only excessive emissions over allocated emissions allowances would be treated as potential liabilities by investors under an ETS). In addition, they find that firms' ability to pass on emissions costs to customers moderates the value-attenuating effects of carbon emissions. Their results suggest that the valuation effect of carbon emissions is a complex issue that is highly dependent on institutional factors such as carbon legislation, the efficiency of the financial market, and legal mechanisms. In line with this argument, Baboukardos (2017) find that the value reduction effects of carbon emissions were alleviated after the passage of the United Kingdom's mandatory carbon report

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regulation. Furthermore, Chapple et al. (2013) and Luo and Tang (2014a) use an event study methodology to find that firms in carbon-intensive sectors suffered a reduction in value upon the release of news that indicated an increased probability of carbon legislation (a carbon tax and ETS) passing in the Australian parliament. In addition, Griffin and Lont (2018) show that the Volkswagen emissions cheating scandal had a substantial and sustained negative effect on the security price of the firm and other large firms in the same industry. However, note that financial markets are inefficient at pricing publicly available information on carbon disclosure and performance because investors can achieve abnormal risk-adjusted returns when simply investing in portfolios constructed on the basis of GHG emissions disclosure and performance.

Overall, the value relevance of carbon emissions is related to the long-term debate over environmental performance and financial performance. Empirical studies show a rather clear trend toward financial market participants becoming more and more sensitive to corporate GHG emissions and managers' response to climate change. The negative relationship between firm carbon emissions and firm value suggests that the market punishes firms for engaging in business as usual and rewards a proactive carbon strategy. These findings imply that firms must balance their goal of economic success with ecosystem stabilisation in an ever more carbon-restrained economy.

The research on capital markets in the carbon accounting literature has at least two limitations. First, current studies largely focus on outcome-based carbon performance (i.e., the level of carbon emissions), with little consideration of other carbon issues. Although this is understandable, because carbon emissions are probably the most readily available data related to firms' carbon management, attempts should be made to obtain other measures of firm carbon performance and test their impact on the capital market. Second, the current literature focuses exclusively on the equity market by considering the effects of firm carbon issues on firm value. Another important capital market—the debt market—is ignored. Although similar research attempts have been identified in other business journals, it is time for accounting scholars to seriously consider this issue. Thus, we suggest that researchers also examine the possible impact of carbon issues on firms' debt financing.

4. Discussion and avenues for future research

Carbon accounting research and practice emerged as an outcome of the Kyoto Protocol and the EU ETS. The increasing awareness of climate change in society and the adoption of accounting methods by organisations to address carbon issues appear to be the main drivers of the growth in carbon accounting research. Although carbon accounting was once treated as a component of CSR or environmental research, the growing literature suggests that it is gradually evolving into an independent research field. Carbon accounting contributes to the

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wider accounting literature by proposing and promoting new accounting practices, such as accounting for carbon assets and liabilities, carbon budgeting, carbon finance, and carbon assurance. These emerging practices and their integration into traditional accounting systems demonstrate the increasingly important role of accounting in sustainable development. Furthermore, the literature shows that firms' carbon decisions are affected by a wide range of external and internal factors, including regulatory pressures, economic pressures, corporate governance, and financial constraints. These studies not only make academic contributions in terms of theory building related to firms' carbon decision making but also have practical implications for policymakers and regulators guiding firms in the transition to a low-carbon economy.

Despite the achievements of the current carbon accounting research, this field is still in the formative stage of development. Although we discuss possible future research throughout the review, here we explore limitations of the current literature and opportunities for future research in more detail.

4.1 Application of theories and theorisation

One shortcoming of the current carbon accounting research is the lack of theorisation around corporate responses to climate change. Instead, the research focuses on applying theories to explain observed phenomena; thus, the main theoretical contribution is the testing of theories. Moreover, the application of theories is dominated by rather broad versions of stakeholder and legitimacy theories (Bebbington & Larrinaga, 2014; Deegan, 2014). Greater theoretical sophistication is needed so that existing practices in carbon accounting can be evaluated and critiqued and new and sounder practices developed (O'Dwyer and Unerman (2016). Therefore, we suggest that carbon accounting researchers adopt new theoretical perspectives to provide nuanced insight into and broaden and deepen understanding of corporate carbon accounting issues.

4.2 Interactions among carbon accounting issues

The four carbon issues (disclosure, assurance, management, and performance) discussed in this literature review do not exist independently of one another. Although there has been some discussion of the link between carbon performance and carbon disclosure (see Section 3.2.3), a link can also be seen between carbon management and carbon performance. For example, the use of target setting as a carbon management tool (Ioannou et al., 2016) or the adoption of a good carbon management system (Tang & Luo, 2014) can have a positive impact on carbon performance. It could be argued that there are even more interactions among the four carbon accounting issues. For example, seeking out carbon assurance may motivate firms to improve their carbon measurement methods and therefore institute better carbon

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management systems. This may in turn improve carbon performance. To date there is limited evidence of interactions among different carbon accounting issues, and this provides opportunities for future research.

4.3 Research on developing countries

The vast majority of papers published in the carbon accounting literature focus on developed countries. We believe that investigating emerging economies may broaden the scope of the research. Not only do some developing countries (e.g., China, India, and Russia) contribute significantly to global emissions, but they also may have distinctive political regimes, less developed financial markets, and different managerial mindsets compared to developed countries. For example, opportunities exist in carbon accounting research in Chinese settings. The country established a national ETS in December 2017 that is expected to surpass the EU ETS to become the largest carbon trading market in the world. This will open the door to a new area of carbon accounting research and practice in the future. Moreover, comparing the results to findings obtained from developed countries can provide additional insights into practical and theoretical issues. Research with a focus on developing countries will complement the current research to provide a more balanced view of global climate change.

4.4 Internal carbon price mechanisms

Internal carbon pricing is becoming a powerful tool for driving reductions in emissions, incentivising low-carbon activities, encouraging innovation, and ensuring sustained competitiveness (World Bank, 2017). In 2017, about 1,400 large companies were factoring an internal carbon price into their business plans, which represented an 8-fold increase from 2014 (CDP, 2017) . Unlike external carbon prices, which are decided by governmental carbon policy (e.g., the carbon tax) or supply and demand (e.g., the carbon trading market), internal carbon prices are determined by organisations themselves. Research can greatly contribute to the academic literature and practice by investigating drivers of internal carbon pricing, methods used to determine internal carbon prices, and ways in which this practice can enhance firm carbon performance and efficiency.

4.5 Green investment and low-carbon initiatives

Green investment and low-carbon initiatives are the centre of a carbon strategy. These initiatives include energy-saving projects, the development of low-carbon products, the use of degradable materials and chemicals, the consumption of renewable energy, and other green projects that can offset carbon and promote energy efficiency. Currently there is a lack of study on the number, nature, scope, and effectiveness of these initiatives. Research in this area can examine expenditures for mitigation, assess company commitment, and evaluate the effectiveness and economic consequences of carbon abatement programs.

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4.6 Carbon accounting and new technology

Technological innovation, such as big data, artificial intelligence, Industry 4.0, cloud technology, digital data, and machine learning, is penetrating traditional accounting information systems. However, current carbon accounting practices do not appear to be in step with the latest technological developments. Researchers may explore the potential of information technology for carbon accounting. For example, Tang and Tang (2019) propose a distributed carbon ledger for integrating carbon asset management and emissions trading based on blockchain technology. We expect that the application of blockchain and other new technology will be a promising area for future carbon accounting research.

5. Conclusion

Experts at the United Nations recently presented comprehensive and overwhelming evidence that the health of the ecosystem is deteriorating more rapidly than ever (United Nations, 2019). Given that climate change is taking on crisis proportions, our safety net is stretched almost to the breaking point, and the current global response is deemed to be insufficient, systematic transformative changes are urgently needed to restore and protect nature (United Nations, 2019). As the replacement of the Kyoto Protocol, the Paris Agreement sends a clear message of the need to decarbonise the economy and encourages businesses to adopt a long-term perspective that balances economy and ecology. In such a context, we perform a state-of-the-art survey of the carbon accounting research. Our findings suggest that carbon accounting did not exist as a field until the introduction of the first ETS in the EU in 2005. Now it is emerging as a distinct discipline and playing a more and more important role in the transition to a carbon-neutral society.

Despite the heterogeneity in the conceptualisation of carbon accounting, there is a consensus that corporate carbon accounting refers to the use of accounting methods to collect, analyse, verify, and report climate change information, account for carbon assets and liabilities, manage carbon risks, and evaluate carbon performance for more informed decision making by managers and external users (Tang, 2017). Our analysis provides a comprehensive and up-to-date picture of historical trends, focal points, and themes in the current carbon accounting literature. The study of carbon accounting is growing rapidly and steadily, in particular after the signing of the Paris Agreement, with a gradual shift from qualitative to empirical studies. Carbon accounting researchers are responsive to climate change and explore and identify various carbon accounting issues that are largely nonexistent in or ignored by previous accounting research. As a result, carbon accounting is emerging as an independent field of study separate from general CSR or sustainability. Carbon accounting research identifies the

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patterns and determinants of managerial responses to climate change as well as the consequences of carbon emissions. It reveals the actions companies take and show how businesses are changing their policies in response to new carbon regulations and legislation, public and stakeholder sentiment, and other factors. It is well recognised that carbon accounting can play a key role in the journey through industry-sponsored decarbonisation programs in the transition to a low-carbon planet.

However, we identify a few gaps in the current literature between the expectations of society and the current state of carbon accounting studies. For example, the current literature is largely descriptive but less constructive. These studies do not devise new technical accounting methods that accountants can adopt to improve carbon accounting practice, detect greenwashing, measure carbon performance, and so on. Furthermore, few novel approaches or unified hybrid accounting–climate schemes are proposed that can guide carbon accounting research and practice. Moreover, the current carbon accounting research is largely descriptive, with no new or innovative theoretical lens. It focuses overwhelmingly on developed nations at the expense of developing ones. Yet firms in some developing countries contribute huge GHG emissions. Moreover, the heavy focus on carbon disclosure is disproportionate to the more direct impacts of carbon management and carbon risk on ultimately achieving carbon reduction. The literature pays less attention to ecological outcomes, such as biodiversity, because it focuses too narrowly on firm value. In addition, there are measurement issues for carbon performance and carbon disclosure quality as well. The literature uncovers a contradiction between corporate carbon disclosure and underlying carbon performance, which resonates with the concern demonstrated in the CSR and sustainability literatures regarding managerial greenwashing. However, it does not provide an effective method of detecting this practice. These gaps between communities' expectations (explicit or implicit) and the status quo of current carbon studies suggest a lack of observable impact on carbon accounting practice. The intensified debate and discussion in the literature does not lead to the creation of formal carbon accounting standards or carbon disclosure guidelines. Moreover, there is a lack of knowledge and understanding of how new carbon assurance standards are implemented in practice, not to mention how these standards should be updated. Therefore, to develop possible generalisable principles for corporate carbon accounting systems, we issue an urgent call to strengthen the role of accounting in improving carbon management systems and strategies.

Carbon accounting and the research face many challenges, and many areas of carbon accounting are not explored in the literature. Thus, we outline here the directions future researchers can consider advancing the study of carbon accounting. First, the research should provide powerful technical tools for carbon accounting, prioritise our research agenda, and develop robust carbon accounting methods for decarbonisation. Novel carbon accounting tools

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and techniques will determine how tomorrow's organisations define, measure, and report their impact on climate change. Second, future research may explore the interaction between carbon accounting and other business systems using a case study approach, which could draw on theoretical frameworks from prior studies (e.g., Burns & Scapens, 2000; Laughlin, 1991). The case study approach can illuminate how decarbonisation becomes infused into day-to-day business practice. Thus, we call for in-depth critical evaluation of carbon accounting practices to reveal the interplay and interactive dynamics of carbon accounting and overall business strategy (Bebbington & Larrinaga, 2014). Currently the knowledge remains speculative, underspecified, and undertheorised. Carbon accounting is expected to play a proactive role in mobilising further complementary and reinforcing changes in the future. This reinforces our endeavour to bolster via research new organisational capacity and competencies that are structured, uniform, and consolidated. To achieve this, accounting must hybridise with other disciplines to find appropriate solutions to these challenging problems. Finally, there is growing demand for carbon accounting and assurance personnel who possess knowledge of climate change and the skills of carbon accounting and assurance. To address the lack of skilled carbon accountants, it is critical that university business schools develop carbon accounting units and textbooks to educate a new generation of accountants.

In sum, we concur with Hopwood (2009), Gray (2002), and O'Dwyer and Unerman (2016) that further high-quality field work is vital to advancing the study of accounting for decarbonisation, which has more or less been neglected (Bebbington & Thomson, 2013). Carbon accounting is expected to provide guidance to policymakers and corporations on best practices in decarbonisation. Accountants can help project carbon costs, prepare carbon budgets and reports, provide assurance, and assist in effective carbon decision making. The current carbon accounting research has provided many insights into the carbon mitigation issues that corporations are facing. It also sends a strong message to firms that, although carbon mitigation can be costly, firms should take early actions that will have a prolonged impact on operational structure, processes, efficiency, reputation, and long-term sustainable outcomes within a carbon-constrained future. The current research findings enhance climate change awareness among corporate managers and provide suggestions for firms evaluating the costs and benefits of their climate change decisions. These findings are valuable for regulators and policymakers in terms of assisting in the design of climate policies and regulations.

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Tables and figures

Table 1. Journals in which carbon accounting papers are published (2005–2018)

No.	ABDC Rating (2016)	Journal Title	ABS AJG 2015 Ranking	Number of Papers	Total
1	**A***	*Accounting, Organizations and Society*	4*	5	
2		*Auditing: A Journal of Practice and Theory*	3	3	
3		*Contemporary Accounting Research*	4	1	
4		*European Accounting Review*	3	4	
5		*The Accounting Review*	4*	2	**15**
6	**A**	*Abacus: A Journal of Accounting, Finance and Business Studies*	3	2	
7		*Accounting and Finance*	2	5	
8		*Accounting and Business Research*	3	2	
9		*Accounting Horizons*	3	2	
10		*Accounting, Auditing and Accountability Journal*	3	18	
11		*British Accounting Review*	3	8	
12		*Critical Perspectives on Accounting*	3	4	
13		*Financial Accountability and Management*	3	1	
14		*International Journal of Auditing*	2	1	
15		*Journal of Accounting and Public Policy*	3	3	
16		*Journal of Accounting, Auditing and Finance*	3	1	
17		*Journal of Business Finance and Accounting*	3	1	
18		*Journal of Contemporary Accounting and Economics*	2	2	
19		*Journal of International Accounting Research*	2	2	
20		*The International Journal of Accounting*	3	4	**56**
21	**B**	*Accounting and the Public Interest*	2	1	
22		*Accounting Forum*	3	6	
23		*Accounting in Europe*	2	2	
24		*Accounting Research Journal*	2	5	
25		*Asian Review of Accounting*	2	1	
26		*Australian Accounting Review*	2	10	
27		*Current Issues in Auditing*	2	1	
28		*International Journal of Accounting and Information Management*	2	1	
29		*Journal of Accounting and Organizational Change*	2	1	
30		*Journal of International Financial Management and Accounting*	2	1	
31		*Managerial Auditing Journal*	2	3	
32		*Pacific Accounting Review*	1	2	
33		*Social and Environmental Accountability Journal*	1	5	
34		*Sustainability Accounting, Management and Policy Journal*	2	7	**46**
	Total			**117**	**117**

Note: ABDC 2016, Australian Business Dean Council; ABS AJG 2015, Association of Business School Academic Journal Guide 2015.

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Table 2. Publications by year

Rating / Year	Number of Publications			
	A*	A	B	Total
2005		1		1
2006				
2007				
2008	3			3
2009	5[a]	2		7
2010		2	4	6
2011		11[b]	4	15
2012		4	7	11
2013		5	8	13
2014	1	3	7	11
2015	2	7	2	11
2016	3	6	4	13
2017	1	8	3	12
2018		7	7	14
Total	**15**	**56**	**46**	**117**

Note:
a – All five papers were published in a special issue of *Accounting, Organizations and Society*.
b – Includes eight papers published in a special issue of *Accounting, Auditing and Accountability Journal*.

Table 3. Publications by research methodology

Rating / Methodology	Number of Publications			
	A*	A	B	Total
Conceptual	5	13	12	**30**
Qualitative empirical	2	11	10	**23**
Quantitative empirical	8	32	24	**64**
Total	**15**	**56**	**46**	**117**

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Table 4. Papers by category (accounting journals, 2005–2018)

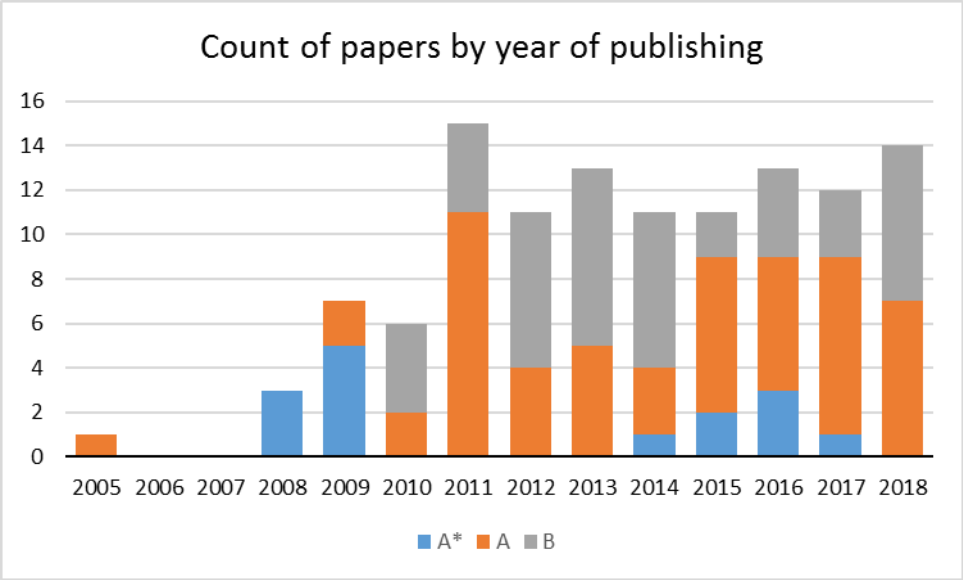
Category	Subcategory	Topic area	Papers
General discussions		Number of papers: 4	Ascui and Lovell (2011), Bebbington and Larrinaga-González (2008), Hartmann et al. (2013), Hopwood (2009)
Policy discussions		7	Andrew and Cortese (2013), Boston and Lempp (2011), Callon (2009), Carter et al. (2011), Lodhia and Martin (2012a), Lodhia and Martin (2012b), McNicholas and Windsor (2011)
Specific corporate carbon issues	Financial accounting for carbon assets and liabilities	9	Black (2013), Cook (2009), Giner (2014), Griffin (2013), Haupt and Ismer (2013), Mete et al. (2010), Raibor and Massoud (2010), Ratnatunga et al. (2011), Warwick and Ng (2012)
	Carbon disclosure	Quality 11	Andrew and Cortese (2011), (Andrew et al., 2010), Comyns and Figge (2015), Cotter et al. (2011), Ferguson et al. (2016), Haigh and Shapiro (2012), Haque and Deegan (2010), Haslam et al. (2014), Kolk et al. (2008), Solomon et al. (2011), Stanny (2018)
		Determinants 28	Alrazi et al. (2016), Choi et al. (2013), Chu et al. (2013), Comyns (2018), Cowan and Deegan (2011), Datt et al. (2019b), de Aguiar and Bebbington (2014), Elsayih et al. (2018), Freedman and Jaggi (2005), Freedman and Park (2014), Freedman and Park (2017), Haque et al. (2016), Hollindale et al. (2019), Hrasky (2012), Liao et al. (2015), Liesen et al. (2015), Liu et al. (2017), Luo (2019), Luo et al. (2012), Luo et al. (2013), Luo and Tang (2014b), Luo and Tang (2016b), Momin et al. (2017), Ott et al. (2017), Rankin et al. (2011), Tang and Luo (2016), Tauringana and Chithambo (2015), Yang and Farley (2016)
	Carbon assurance	13	Datt et al. (2018), Datt et al. (2019a), Green and Li (2012), Green and Taylor (2013), Green and Zhou (2013), Huggins et al. (2011), Kim et al. (2016), Martinov-Bennie (2012), Martinov-Bennie and Hoffman (2012), Olson (2010), Simnett et al. (2009), Tang (2019), Zhou et al. (2016)
	Carbon management	Practices 11	Bui and Fowler (2019), Burritt et al. (2011), Deegan and Islam (2012), Engels (2009), ,Ioannou et al. (2016), Lee and Wu (2014), Linnenluecke et al. (2015), Ratnatunga

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			and Balachandran (2009), Tang and Luo (2014), Vesty et al. (2015)
		Determinants 11	Bui and de Villiers (2017), Bui et al. (2020), Cadez and Guilding (2017), Herbohn et al. (2012), Jeffrey and Perkins (2014), Kumarasiri and Gunasekarage (2017), Kumarasiri and Jubb (2016), Luo and Tang (2016a), Moore and McPhail (2016), Trotman and Trotman (2015), Yunus et al. (2016)
	Carbon performance	Measurement 5	MacKenzie (2009), Bowen and Wittneben (2011), Cooper and Pearce (2011), Milne and Grubnic (2011), Young (2010)
		Determinants 6	Birchall et al. (2015), Broadstock et al. (2018), Haque (2017), Hassan and Kouhy (2013), Jeffrey and Perkins (2015), Qian and Schaltegger (2017)
Impact on the capital market		12 **Total: 117**	Baboukardos (2017), Chapple et al. (2013), Clarkson et al. (2015), Cooper et al. (2018), Griffin and Lont (2018), Griffin and Sun (2013), Griffin et al. (2017), Johnston et al. (2008), Liesen et al. (2017), Luo and Tang (2014a), Matsumura et al. (2014), Saka and Oshika (2014)

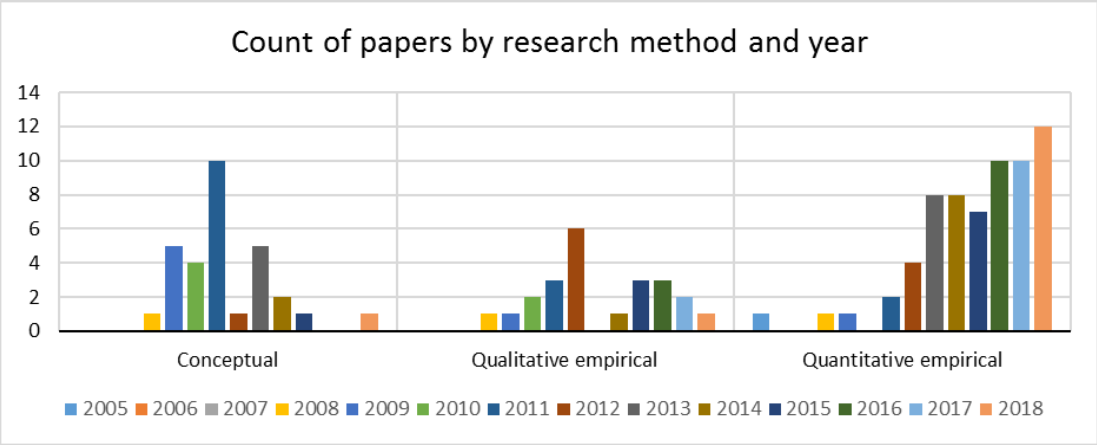
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Figure 1. Number of publications by year



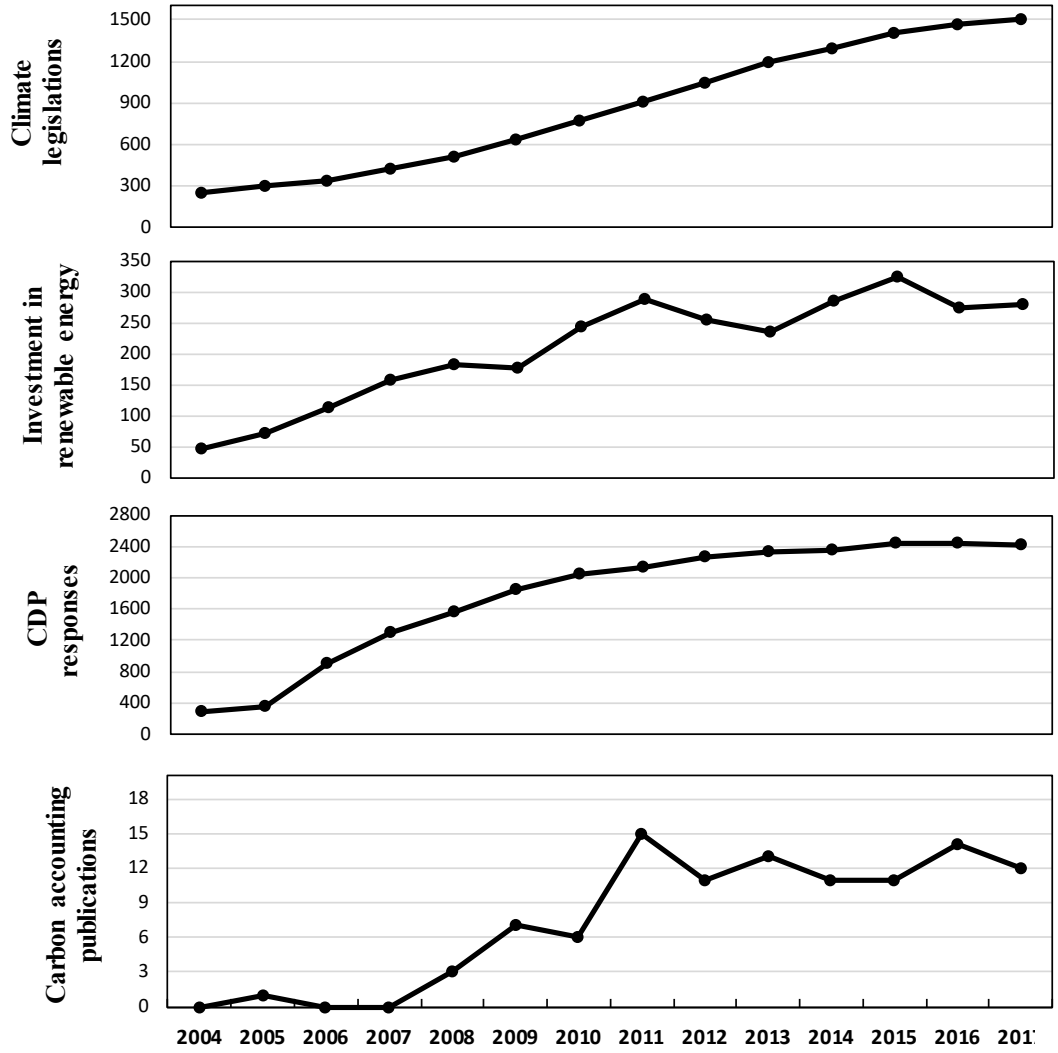
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Figure 2. Distribution of publications by research methodology



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Figure 3. Synchronisation of responses to climate change



Data sources:

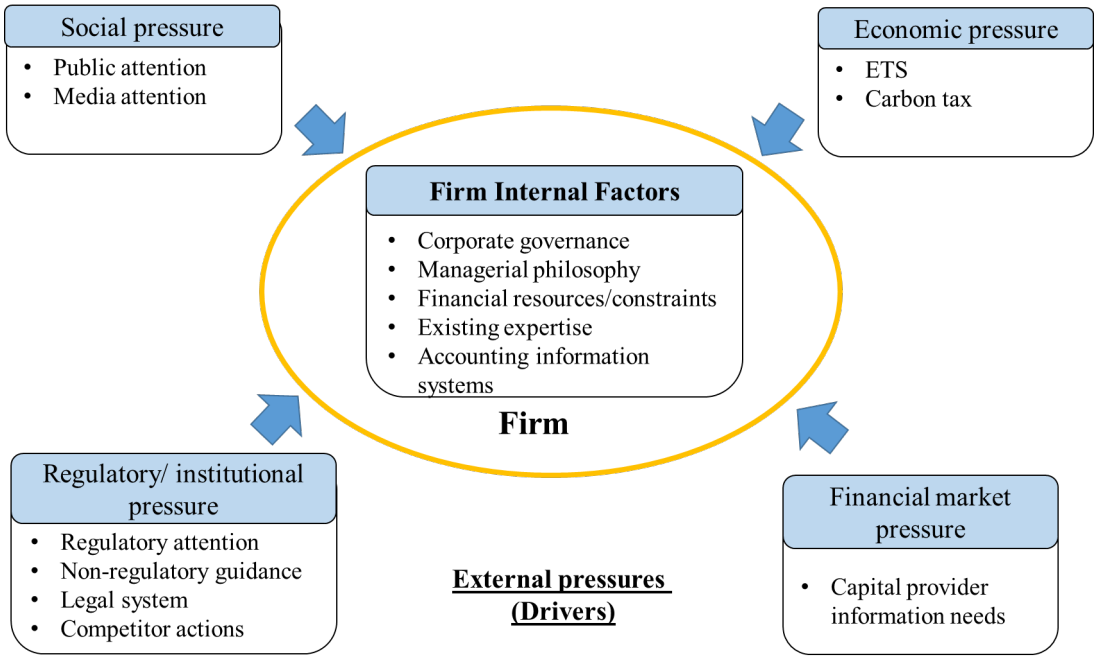
- Climate legislation:
  The Grantham Research Institute on Climate Change and the Environment

  http://www.lse.ac.uk/GranthamInstitute/climate-change-laws-of-the-world/?region=all&country=all&fromyear=all&toyear=all&emitter=all&income=all&framework=all&execleg=all&category=all&type=law
- Investment in renewable energy:
  UN Environment's Economy Division, Frankfurt School-UNEP Collaborating Centre for Climate & Sustainable Energy Finance, and Bloomberg New Energy Finance, 2018, Global trends in renewable energy investment 2018

  http://www.greengrowthknowledge.org/resource/global-trends-renewable-energy-investment-report-2018
- CDP responses: CDP database
- Carbon accounting publications: this study

Electronic copy available at: https://ssrn.com/abstract=3947433

Figure 4. A framework of the determinants of corporate carbon disclosure, adopted from Luo et al. (2013). ETS, emissions trading scheme.



Electronic copy available at: https://ssrn.com/abstract=3947433